

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EC-804-GS
B.E. VIII Semester
Examination, December 2020
Grading System (GS)
TV and Radar Engineering
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) With the help of block diagram explain the basic principle of sound and picture transmission.
b) Explain flicker. How it can be removed?
2. a) Discuss about the consolidated CCIR system - B standards.
b) Explain the working principle of Videocon camera.
3. Explain working of digital TV transmitter in brief.
4. Explain the following :
 - i) Aspect ratio
 - ii) Interlaced scanning
 - iii) Picture and sound modulation
5. a) Explain briefly about HF over horizontal radar.
b) Discuss the principle working of Bistatic radar.

[2]

6. a) Explain the working of colour television camera.
b) Explain the working of Trinitron picture tube.
7. Draw and explain the block diagram of PAL encoder and decoder.
8. a) Derive expression for the Radar range equation. What is the role of integration of pulses in the Radar range equation.
b) Calculate the maximum range of Radar in which operate at 3cm with a perk pulse power of 500 kW, its minimum receivable power is 10^{-13} W, the capture area of its antenna is 5m^2 and target cross section area is of 20m^2 .
